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IOT-BASED AQI METER SYSTEM for Real-Time Data Analysis and Decision Making

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ABSTRACT

- ✓ **OVERVIEW** Development of a system for measuring real-time indoor and outdoor air quality.
- ✓ **GOAL** Facilitate better decisions through data analysis.
- ✓ **COMPONENTS** In different sectors, the application of sensors empowered by IoT and their accompanying cloud storage facilities has been growing tremendously.

PROBLEM STATEMENT

- **ISSUE** Observe aspects of air pollution which are harmful to health and all living organisms, chiefly because of urbanization and industrialization.
- **FOCUS** Identify and study the effects of pollutants and provide for their continuous monitoring.

OBJECTIVES

➤ Real-time IoT-based air quality monitoring system would develop.

➤ **KEY GOALS**

- Implement cloud-based analytics.
- Evaluate resource effectiveness, precision, and operational ease.
- Identify entrepreneurial opportunities for commercialization.

PROPOSED SOLUTIONS

- **HARDWARE COMPONENTS**

Arduino, ESP32, ESP8266, MQ series gas sensors, PM2.5 and PM10 sensors.

- **SOFTWARE COMPONENTS** Blynk App, Firebase, Android Studio, MATLAB, Python, etc.

REAL-TIME MONITORING

- **Functionality** User-friendly data processing and instantaneous display.
- **AQI Calculation** Based on standardized environmental parameters.

ALERTS AND NOTIFICATION

- **Feature** Threshold alerts for hazardous air quality levels.
- **Purpose** Inform users about air quality risks.

MANAGEMENT AND ENTREPRENEURSHIP SCOPE

- ✓ **Market Growth** Increasing awareness about pollution.
- ✓ **Target Customers** Households, companies, government institutions, healthcare.
- ✓ **Revenue Model** Device sales, subscription services, custom solutions.

BENEFIT AND IMPACT

- ✓ **Advantages:**
- ✓ Improved air quality.
- ✓ Reduced exposure to pollution.
- ✓ Healthier lifestyle promotion.
- ✓ Economic growth and healthcare cost savings.

CONCLUSION

- The IoT-based AQI Monitoring System offers a low-cost, scalable solution for air quality monitoring.
- Enhances real-time data use, contributing to a cleaner environment and business opportunities.